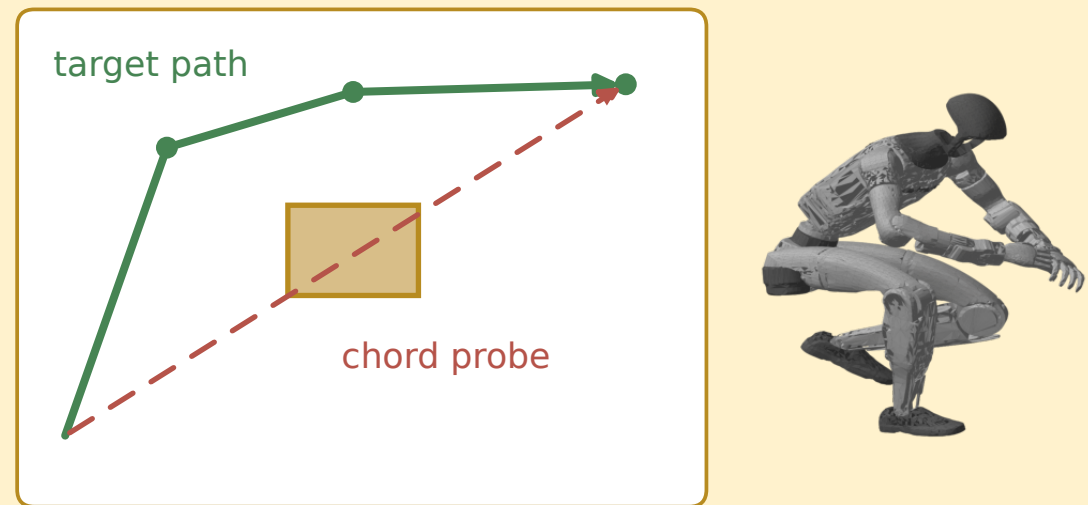


Motion2Scene: from motion evidence to scene-and-goal control

Three linked learning problems: infer useful geometry, preserve movement capability, and generate commands from task context.

(a) Learn scenes from motion

Full motion → event-conditioned proposer



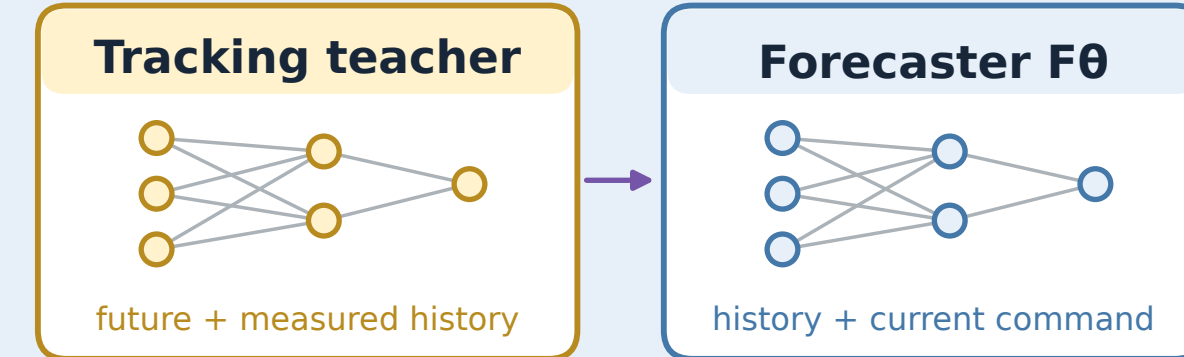
Hindsight: preserve the observed movement; place geometry that challenges an alternative.

225 fixed recipes → learned categorical scores

Clearance mask → sequential scene sampling

Schematic geometry; reference pose, not a rollout.

(b) Recover the full-command motor



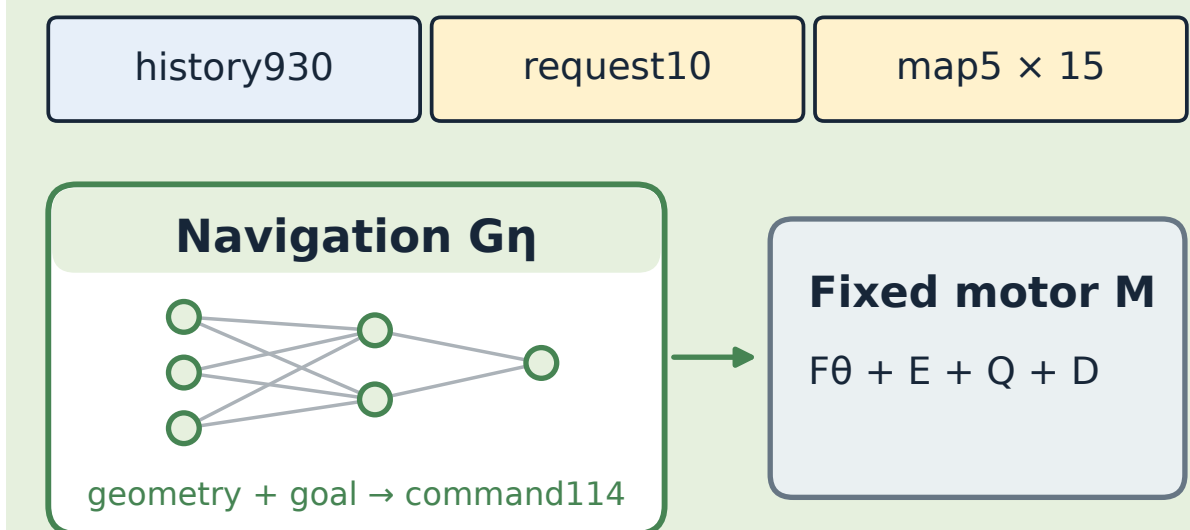
Online action / token distillation

Predicted reference → fixed E → Q → D

Retained motor representation → 29D action

Teacher recovery and scene fitting are separate studies.

(c) Learn navigation from context



Full command is completed internally.

Approach → brake → stable goal hold

Known-map control; no reference at inference

Camera context and wider tasks remain research branches.

(d) Convert compatible scenes and motor capability into qualified supervision

Scene + goal → execute continuation → score task → admit supported rows

Fresh 50-tick hold; distance, speed, contact, fall and deadline. Retain failed attempts and collection cost.

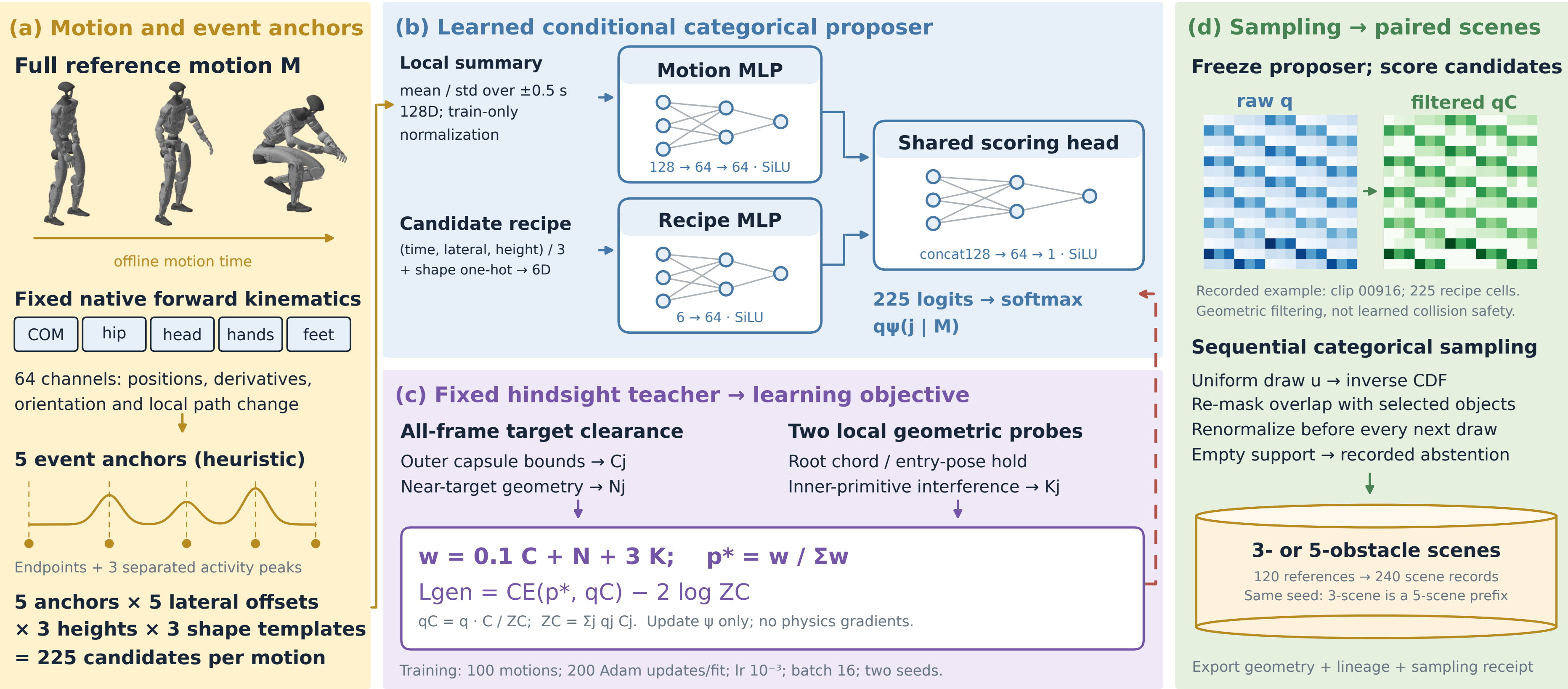
Learner-state recovery supplies bounded local support; scene acceptance alone supplies no action label.

Fit navigation; roll out again

Current evidence: 4/8 initial tasks,
1/8 on evaluation-seed confirmation.
End-to-end data utility remains unproven.

Motion2Scene: learning a hindsight obstacle distribution

Dataset-generating branch • 21,185 learned parameters • geometry-derived supervision • zero dynamics in generator fitting



(e) Why this is hindsight — and where physical teaching begins



Motion supplies its own geometric supervision.

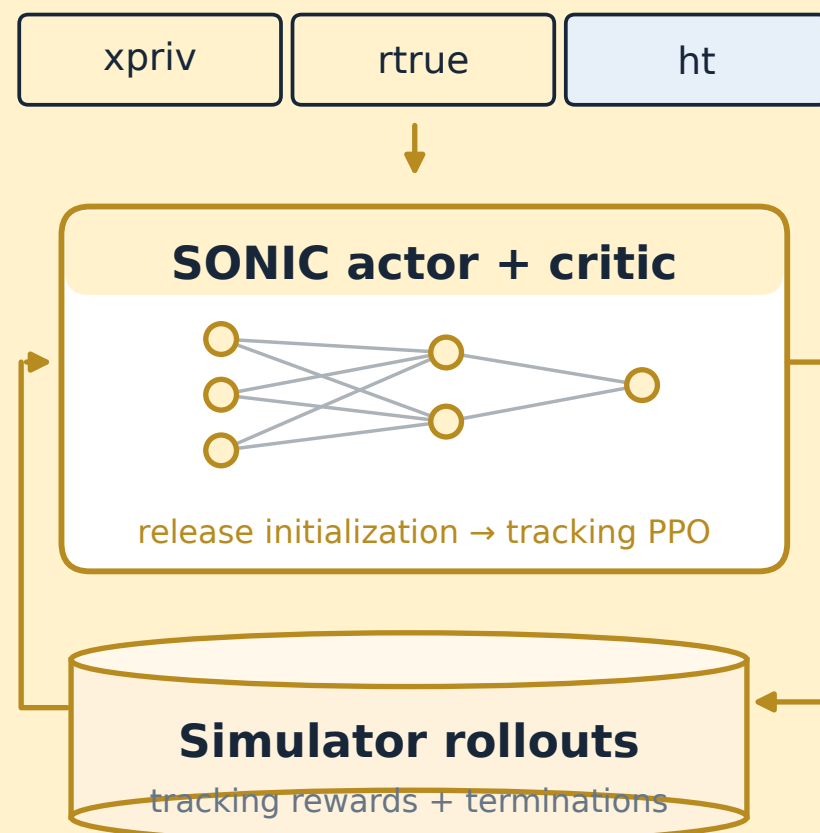
Offline access to the complete motion allows hindsight labels.
Learn preference over fixed recipes, not a unique original room.
Chord / held-pose probes require separate physical qualification.

Separate physical qualification

Repair reference → execute compatible scene continuation
Measure goal hold, contact, fall, and termination
Only executed task evidence supplies positive action labels

Current proposed full-command architecture: predict missing desired-reference context while preserving the pretrained SONIC motor representation.

Repaired G1 motion library



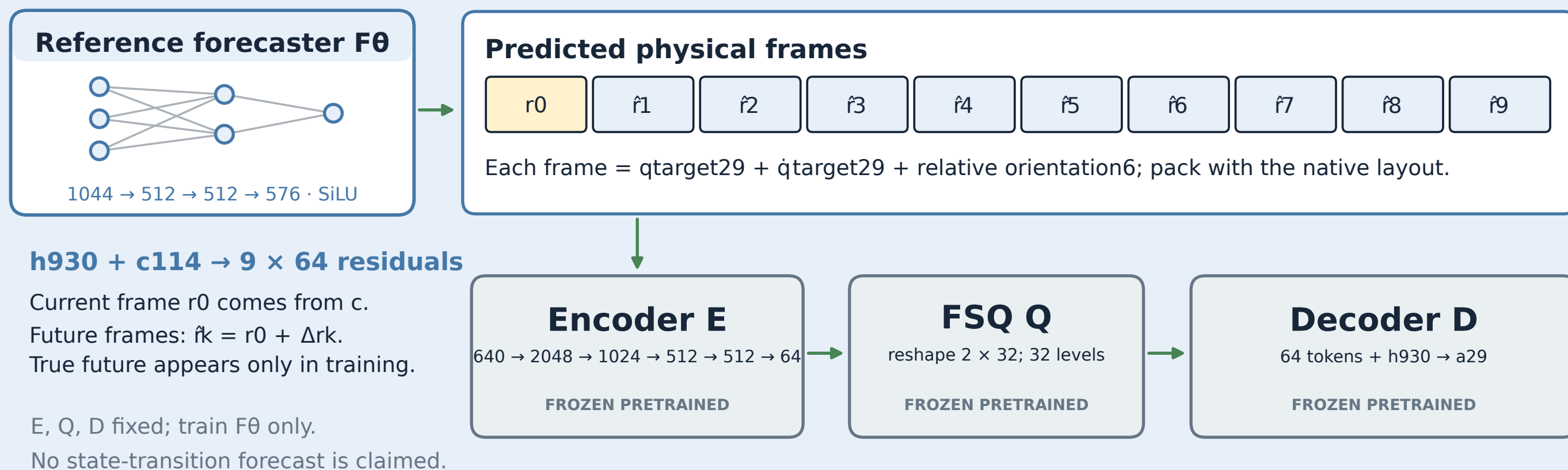
(b) Public full-command interface — 114 values, 113 available

heading 2	velocity 3	height 1	yaw rate 1	arrival 1	keypoints 42	joints 29	joint vel. 29	orientation 6
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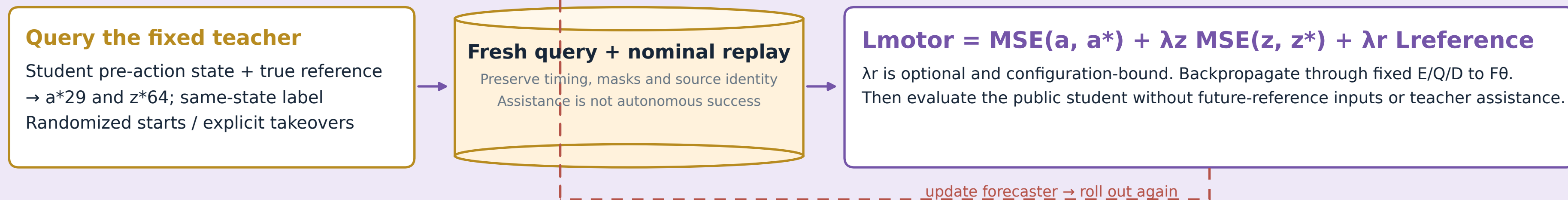
Current target only + native measured history $h \in \mathbb{R}^{930}$. Arrival coordinate 7 stays masked.

The 79D masked CVAE and 114D transformer are earlier controls; the recovered specialist below is a deterministic reference forecaster.

(c) Full-command inference through the retained motor path



(d) Online distillation — teacher actions at the states visited by the student

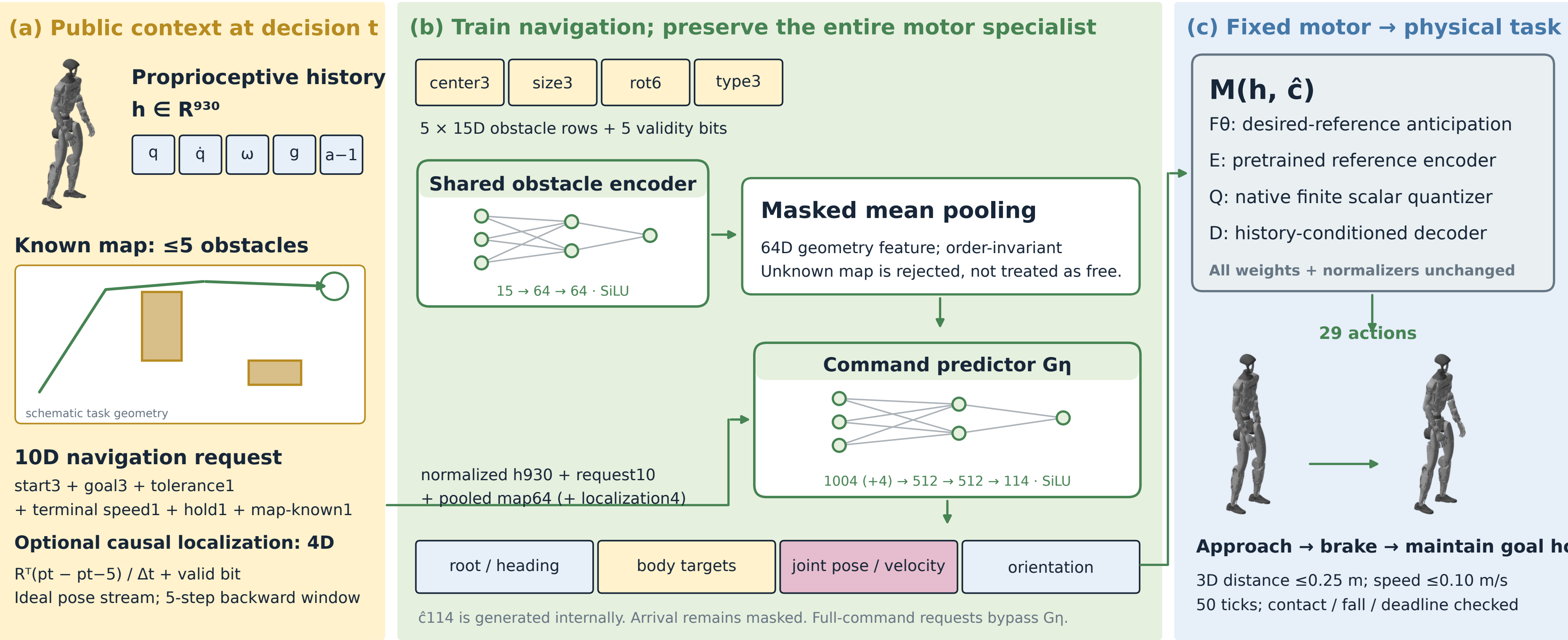


Green: action inference. Amber: privileged training inputs. Red dashed: gradient/update path. Gray: frozen components. Reference pose renders are illustrative.

Do not identify this deterministic motor specialist with BFM's CVAE or BFM-Zero's forward-backward representation learning.

From full commands to scene-and-goal behavior

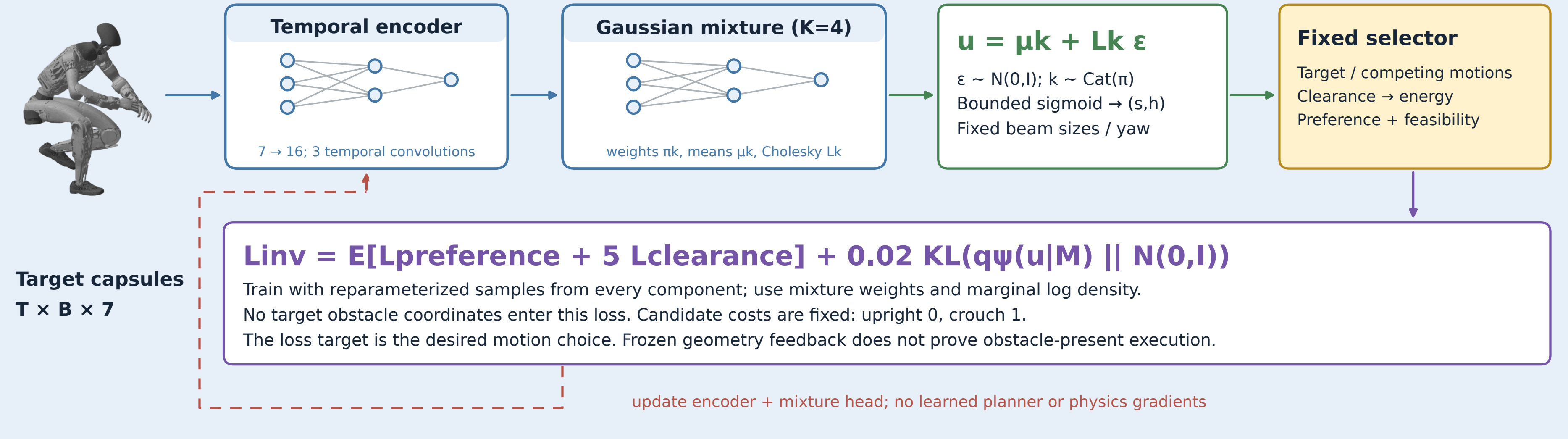
The navigation actor completes a rich motor command internally; the public interface requires no motion ID, reference phase, or external reference trajectory.



A separate Motion2Scene branch: differentiable inverse beam learning

This Gaussian beam diagnostic predates the categorical dataset generator. Its losses and sampling law must not be attributed to hindsight-801.

(a) Motion-only inference; fixed geometric preference supplies training feedback



Dataset generator (Figure 2): fixed discrete recipes + geometric target distribution + coverage/acceptance loss.

This diagnostic: continuous station/height samples + fixed motion-choice energy + target barrier + distribution KL.